Assignment prep: pool heating system

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Complete the system diagram to meet the specification. Show the external input and output clearly.

- Input: set (desired) temperature
- Output: heat (water temperature)

2 Complete the sub-system diagram to meet the specification. Show the external input and output, all sub-systems, the system boundary and the interactions between sub-systems. Any output device needs its own driver.

- External input: set temperature into the microcontroller
- Microcontroller → driver → heater
- External output: heat into the water
- Temperature sensor measuring the water, with a feedback arrow to the microcontroller
- Dashed system boundary round the microcontroller, driver, heater and sensor
- Arrows show every interaction between sub-systems